WORKED SOLUTION

STEP II 2022, Question 11

Probability and Statistics • Expected Loss Optimisation

STEP II 2022, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Split the story by what happens to the T tested sticks. If any of them is infected, nothing goes on the network and the loss is zero; only when all T are clean does the network gamble come into play. Writing the expected loss as a function of q turns (ii) into a one-variable maximisation, and comparing consecutive values of T (a difference, not a derivative, since T is an integer) handles (iii).

Solution

Throughout, a loss of $-x$ means a gain of £x. Let L be the net loss in pounds. Sticks are infected independently, each clean with probability q, so any particular set of m sticks is entirely clean with probability q^m. Which T sticks are chosen doesn't matter, by symmetry.

Part (i)

Take $T > 0$ first. There are three outcomes.

- Some tested stick is infected (probability $1 - q^T$): the rest are destroyed, nothing is used, and $L = 0$.
- All T tested sticks are clean, but at least one of the other $N - T$ is infected (probability $q^T(1 - q^{N-T})$): we pay for disinfection, so $L = D$.
- All N sticks are clean (probability $q^T q^{N-T} = q^N$): we gain £1 on each of the $N - T$ sticks used, so $L = -(N - T)$.

Hence

$$\begin{aligned} \mathrm{E}(L) &= D\,q^T\left(1 - q^{N-T}\right) - (N - T)\,q^N \\ &= D\,q^T - (N - T + D)\,q^N. \end{aligned}$$

When $T = 0$ all N sticks are used: $L = D$ with probability $1 - q^N$ and $L = -N$ with probability q^N, giving $D - (N + D)q^N$. That is the same formula with $T = 0$, so for every $0 \leq T < N$

$$\mathrm{E}(L) = D\,q^T - (N - T + D)\,q^N.$$

Part (ii)

Bounds on α. Since $D > 0$, $T \geq 0$ and $N - T > 0$, numerator and denominator satisfy $DT \geq 0$ and $N(N - T + D) > 0$, so $\alpha \geq 0$. Also

$$N(N - T + D) - DT = N(N - T) + D(N - T) = (N - T)(N + D) > 0,$$

so $DT < N(N - T + D)$ and $\alpha < 1$.

The maximum. Write $f(q) = Dq^T - (N - T + D)q^N$ for $0 \leq q \leq 1$, with N, D, T fixed.

If $T = 0$ then $\alpha = 0$ and $f(q) = D - (N + D)q^N$, which decreases in q, so the greatest value is at $q = 0$, i.e. where $q^N = 0 = \alpha$.

If $T \geq 1$, differentiate:

$$f'(q) = DTq^{T-1} - N(N - T + D)q^{N-1} = N(N - T + D)\,q^{T-1}\big(\alpha - q^{N-T}\big).$$

> **Ask yourself** How do we know this stationary point is a maximum, and the greatest value on the whole interval?

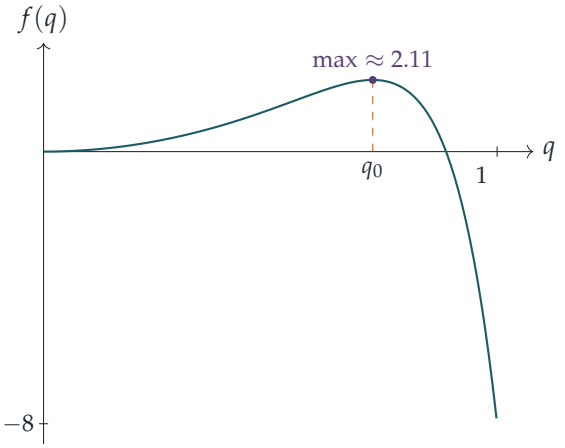
The factor $N(N - T + D)q^{T-1}$ is positive for $0 < q \leq 1$. Because $N - T \geq 1$, q^{N-T} is strictly increasing from 0 to 1, and $0 < \alpha < 1$, so there is exactly one $q_0 \in (0, 1)$ with $q_0^{N-T} = \alpha$. For $q < q_0$ we have $f'(q) > 0$ and for $q > q_0$ we have $f'(q) < 0$. So f rises up to q_0 and falls afterwards: its greatest value on $[0, 1]$ occurs where $q^{N-T} = \alpha$, as required.

The greatest value. At q_0 we have $q_0^{N-T} = \alpha$, so $q_0 = \alpha^{1/(N-T)}$ and $q_0^T = \alpha^{T/(N-T)} = \alpha^k$. Then

$$\begin{aligned} f(q_0) &= q_0^T\big(D - (N - T + D)\,q_0^{N-T}\big) = \alpha^k\Big(D - (N - T + D)\cdot\frac{DT}{N(N - T + D)}\Big) \\ &= \alpha^k\Big(D - \frac{DT}{N}\Big) = \frac{D(N - T)\,\alpha^k}{N}. \end{aligned}$$

So the greatest expected net loss is £$D(N - T)\alpha^k/N$. (For $T = 0$ this reads £D, matching $f(0) = D$ if we take $\alpha^0 = 1$.)

The graph shows f for $N = 10$, $T = 2$, $D = 5$, where $\alpha = \frac{1}{13}$.



Part (iii)

Now fix N, D and q, and write $E_T = Dq^T - (N - T + D)q^N$. Since T only takes integer values, "increasing in T" means $E_{T+1} > E_T$ for $0 \leq T \leq N - 2$. Subtracting,

$$\begin{aligned} E_{T+1} - E_T &= Dq^{T+1} - Dq^T + q^N \\ &= q^T(q^{N-T} - Dp), \end{aligned}$$

using $q - 1 = -p$.

Ask yourself The bracket depends on T. How can one β work for every T at once?

Because $0 < q \leq 1$ and $N - T \leq N$, we have $q^{N-T} \geq q^N$, so it is enough to make $q^N > Dp$, a condition that doesn't involve T. By Bernoulli's inequality (an easy induction: if $(1-p)^n \geq 1 - np$ then $(1-p)^{n+1} \geq (1-np)(1-p) = 1 - (n+1)p + np^2$),

$$q^N = (1-p)^N \geq 1 - Np.$$

So if $1 - Np > Dp$, that is $p < \dfrac{1}{N+D}$, then $q^{N-T} \geq q^N \geq 1 - Np > Dp$, and also $q^T > 0$. Therefore, with

$$\beta = \frac{1}{N+D} > 0,$$

every $p < \beta$ gives $E_{T+1} > E_T$ for all T, so the expected loss is an increasing function of T.

For such p the smallest of $E_0, E_1, \ldots, E_{N-1}$ is E_0. So when p is small enough, the best policy is $T = 0$: test nothing and use all N sticks. This makes sense: if infections are rare, testing mostly destroys good sticks that would have earned money.

Checks

- With $N = 1$, $T = 0$ the formula gives $D - (1 + D)q$: lose D with probability p and gain 1 with probability q, as expected.
- $q = 1$ (no infection at all) gives $D - (N - T + D) = -(N - T)$, a sure gain of $N - T$.
- For $N = 10$, $T = 2$, $D = 5$: $\alpha = \frac{1}{13}$, $k = \frac{1}{4}$, and the maximum is $4 \cdot 13^{-1/4} \approx 2.107$, matching the peak of the graph above. A computer check confirmed the part (i) formula by listing every infection pattern for $N \leq 7$.
- For $N = 10$, $D = 5$, $p = 0.3$ (well above $\beta = \frac{1}{15}$), the expected loss actually *decreases* with T, from about 4.58 at $T = 0$ to 0.03 at $T = 9$: so some condition on p is genuinely needed.

Common slips

- Forgetting the outcome "a tested stick is infected", which contributes 0 to the expectation but still uses up probability $1 - q^T$.
- Counting the gain as £N instead of £$(N - T)$: the tested sticks are destroyed even when they are clean.
- In (iii), differentiating with respect to the integer T without comment, or choosing a β that depends on T.

Beyond STEP

This is a small problem in statistical decision theory: choose an action (here T) to minimise expected loss when a parameter (p) is unknown. Part (ii) is a *minimax* idea: it finds the worst case over all possible p for each policy, which is what a cautious manager might compare. The same trade-off between the cost of testing and the cost of missed defects lies behind acceptance sampling and group testing. Blitzstein and Hwang, *Introduction to Probability*, is a good place to build the expectation toolkit used here.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.